www.tntextbooks.in

Microscopic measurements:

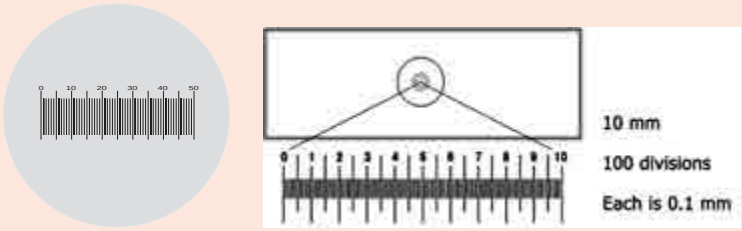
The microscope also has facility to measure microscopic objects through a technique called '**micrometry**'. There are two scales involved for measuring.

1. Ocular Micrometer
2. Stage Micrometer

Ocular Micrometer: It is fixed inside the eye piece lens. It is a thin transparent glass disc where there are lines divided into 100 equal units. The scale has no value.

Stage Micrometer: This is a slide with a line divided into 100 units. The line is about 1mm. The distance between two adjacent lines is 10 µm. The known value of the stage micrometer is transferred to the ocular micrometer, thereby the measurements can be made using ocular micrometer.

$$\text{The distance between two adjacent line of ocular meter} = \frac{\text{Number of stage divisions}}{\text{Number of ocular divisions}} \times 10$$



(a) (b)

Figure 6.3: a. Ocular micrometer; b. Stage micrometer

Light passes with different velocity after coming out of the thinnest and thickest areas of the phase plate thereby increasing the contrast of the specimen. A hollow cone of light passes through the condenser. Direct light pass through thin area of phase plate, whereas light passing from the specimen reaches thick area of phase plate. The light passing through thicker area travel at low speed, on the other hand the light passing through thin area reach fast therefore contrast is increased in the specimen. Phase contrast microscope is used to observe living cells, tissues and the cells cultured *invitro* during mitosis (Figure 6.2 f and g).

6.2.4 Electron Microscope

Electron Microscope was first introduced by **Ernest Ruska** (1931) and developed by **G Binning** and **H Roher** (1981). It is used to analyse the fine details of the cell and organelles called ultrastructure. It uses beam of accelerated electrons as source of illumination and therefore the resolving power is 1,00,000 times than that of light microscope.

The specimen to be viewed under electron microscope is dehydrated and impregnated with electron opaque chemicals like gold or palladium. This is essential for withstanding electrons and